KeroshiZeroTetris:
Self-Learning Modern Tetris AI

A Preprint

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May 2026

1 Introduction

Traditional Tetris is a classic single-player survival game, and Modern Tetris[1] introduces something new. Modern Tetris is usually played in a 1v1 competitive mode, and features an attack system. Players can perform specific clears to send garbage lines to their opponents. In addition, Modern Tetris also introduces the Super Rotation System (SRS), so players can perform clears like T-Spin to gain more attack.

(a) A T-Spin Double.

(b) A T-Spin Triple.

Figure 1: Some sample clears with SRS.

In Modern Tetris, human players often try to perform higher Attack Per Minute (APM), but Tetris AI needs to perform higher Attack Per Piece (APP).

To challenge the highest APP, we present KeroshiZeroTetris based on AlphaZero[1].

However, directly applying the AlphaZero framework to Modern Tetris brings some difficulties:

- Unlike chess or Go, the next state is not fixed because of the 7-Bag random generator.
- The neural network for Modern Tetris must be carefully designed to predict both policy and value information.
- The Tetris AI needs to play indefinitely with a high APP, so a major challenge is how to set up training tasks and how to implement the trained policy in actual match play.

To solve these issues, this paper details the implementation of KeroshiZeroTetris.

[1] Examples include competitive platforms like TETR.IO and Techmino, as well as official games like Tetris Effect: Connected.

References

[1] David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and go through self-play. Science, 362(6419):1140–1144, 2018.